

Convolutional Neural Networks for Automated Cervical Cancer Detection: A Comprehensive Literature Review

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Abstract—Cervical cancer is a major in world health challenge especially in the low and middle income countries whereby inadequate screening is the cause of high mortalities with traditional tools such as the Pap smears being impaired by subjective results, cell overlap and the lack of resources that justifies the use of automated deep learning technologies to address the problem. The paper presents a review of the convolutional neural networks (CNNs), transfer learning, ensembles and Vision Transformers in detecting cervical cancer in cytological, histopathological and MRI images with more than 95 percent accuracy on such datasets as SIPaKMeD. It introduces a new attention-enhanced CNN to multiclassify cervical histopathology on 25,000 RGB images with 5 classes (dyk, koc, mep, pab, sfi) in stratified and graded 70-15-15 baseline CNN with Batch Normalization, ReLU and Max-Pooling and augmented with Convolutional Block Attention Module (CBAM) to refine channels and spatial locations and Image Baseline transfer models such as the ResNet50 and ResNet101 also exhibit perfect class-wise precision, recall and F1-score of 1.00, and AUC=1.00, which was better than the DenseNet201, and the lightweight CNN proposed exhibits steady convergence and no overfitting, and also it is clinically validated and the proposed focus on pathological markers which supports a WHO goal of cervical cancer elimination.

Index Terms—cancer, cervical cancer, CNN

I. INTRODUCTION

Cervical cancer is one of the leading global challenges that are related to public health as it is considered as the fourth most common cancer in women that can be found in the world. In 2022, some 660,000 new cases were observed, and this led to the deaths of about 350,000 of them, of which more than 94 percent of them were found in low- and middle-income countries where the availability of screening and treatment is suboptimal. Despite the fact that, to a considerable extent, HPV vaccination and early diagnosis would prevent this disease, healthcare disparities remain one of the primary factors contributing to high mortality rates particularly in sub-Saharan Africa, Central America, and South-East Asia. When the disease is detected at precancerous or early stages, the survival rates are increased dramatically, thus it can be highly cured. Conventional screening procedures such as Pap smears and HPV have decreased deaths in the high-resource setting, but are challenged with some drawbacks such as subjective

reporting, lack of pathologists, overlapping of cells, poor contrast of the image, and overworked with scarce resources in the limited-resource setting mostly generate errors and can cause delays [3, 8]. Recent progress in deep learning, especially convolutional neural networks, transfer learning, ensembles and convolutional neural networks with Vision Transformers demonstrated tremendous potential to detect cervical cancer in cytological images, histopathology, and MRI segmentation, fully automated [2, 5, 7, 9, 10]. The current paper presents a detailed literature review of deep learning-based cervical cancer detection, outlining the main developments, model results, databases, and difficulties. It is intended to point out how automated systems can enhance screening equity and make progress in the mission of the World Health Organization to eradicate cervical cancer as a public health issue.

II. LITERATURE REVIEW

Conventional neural networks (CNNs) and other deep learning approaches have profoundly facilitated the process of automated detection of cervical cancer based on cytological such as Pap smears and ThinPrep cytologic test (TCT)) and histopathological (data) cytology. Overlapping cells, low contrast, and the necessity of an effective screening to decrease the work of pathologists are some of the issues that these approaches deal with [3].

A. General and Multi-Cancer Applications

Most studies consider cervical cancer but transfer learning models such as DenseNet121 have shown excellent levels of multi-cancer histopathology classification whereby, in datasets with brain, breast, and kidney tumors, transfer learning has provided high rates of validation of 99.94 percent following the application of Otsu binarization and watershed segmentation [1].

B. Studies and Challenges Review Process

In-depth surveys note the place of the feature extraction and the models such as SVM and CNNs, particularly, in the classification of Pap smears with some challenges such as cell overlap and low contrast, and the workload could be more efficient in the automation process [3]. Segmentation is a technique to detect cervical cancer based on MRI which has

moved away traditional segmentation (thresholding, region-growing) to the deep learning architecture, such as U-Net, V-Net, and DeepLab, with an improved sensitivity to detect cervical cancer at an early stage [10].

C. Transfer Learning and Individual CNN Models

CNNs used in the analysis of Pap smears are trained with transfer learning. Finetuning on GoogleNet demonstrated 96.01 percent accuracy which is better than ResNet-50 used as feature extractor and got 92.03 percent accuracy[2]. R-CNN models trained on TCTs images across several hospitals had a sensitivity of 99.4 percent with a specificity of 34.8 percent on lesion samples and stage classification [8]. Scratch trained Vision Transformers (ViT) achieved 94 percent accuracy, and hybrid CNN-ViT models including HViT-Cerv models improved to 97.99 percent [5].

D. Techniques of Ensemble and Advanced

Ensemble techniques increase reliability. The accuracy rate of a fuzzy rank-based ensemble of Inception v3, Xception, and DenseNet-169 based on the SIPaKMeD and the Mendeley LBC datasets was 98.55 and 99.23 percent, respectively, through the integration of confidence-weighted decisions [9]. Multi-deep transfer learning using feature extraction in models such as MobileNet, DenseNet, EfficientNet, Xception, and ResNet50, and PCA to reduce features and a new smoothing cross-entropy loss functional reached 97 percent accuracy on SIPaKMeD dataset [7].

E. Non-Image-Based Approaches

To predict risks based on clinical factors with deep CNNs, there was a higher level of accuracy of clinical factors reaching around 90 percent, at the expense of both SVM and the Random Forest [6]. A web-based system, that combined several machine learning models (ANN, AdaBoost, KNN, RFC, SVM, and Decision Trees) with biopsy data on 858 patients, demonstrated that RFC and Decision Trees were the most effective in the early-detection of how many patients might have [4]. All in all, deep learning models always obtain high accuracies (more than 95 percent in most cases), and ensembles, hybrids, and sophisticated loss functions have better performance [5, 7, 9]. The work of the future could be able to combine multi-modal data analysis like MRI, with cytology to do a detailed diagnosis [10].

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III. METHODOLOGY

The proposed research methodology follows a multi-stage computational pipeline designed to achieve high-fidelity classification of cervical cancer histology.

A. Dataset Description

The study utilizes the Multi-Cancer Dataset (Cervical subset) sourced from Kaggle, containing 25,000 high-resolution RGB images categorized into five pathological

classes: *cervix_dyk*, *cervix_koc*, *cervix_mep*, *cervix_pab*, and *cervix_sfi*.

- **Data Partitioning:** The dataset was partitioned using a stratified split into training (70%), validation (15%), and testing (15%) subsets.
- **Normalization:** Images were resized to 224×224 pixels and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

B. CNN Architecture

The baseline model consists of four convolutional blocks. Each block employs a 3×3 kernel, Batch Normalization, ReLU activation, and a 2×2 Max-Pooling layer. The filter depth progresses through 32, 64, 128, and 256 to capture hierarchical morphological features.

C. Attention-Enhanced Model (Proposed)

The proposed architecture integrates the Convolutional Block Attention Module (CBAM) into the CNN backbone. CBAM refines feature maps through two sequential sub-modules: Channel Attention, which identifies significant feature detectors, and Spatial Attention, which identifies salient biological regions within the cell.

D. Model Interpretability via Grad-CAM

To ensure clinical transparency, Gradient-weighted Class Activation Mapping (Grad-CAM) is implemented. This technique produces heatmaps highlighting the regions of interest (ROI) that influenced the model's prediction, ensuring focus on pathological markers.

IV. THEORETICAL FRAMEWORK

A. Dataset Theory

Histopathological analysis of cervical cells is rooted in the detection of microscopic morphological markers. Theoretically, RGB images are required to capture *nuclear atypia*—defined as pathological variations in nuclear size and chromatin distribution. A balanced distribution of 5,000 images per class provides a safeguard against majority class bias.

B. CNN Theory

The feature extraction is based on hierarchical learning. The convolution operation between an input X and a filter W is mathematically expressed as:

$$Y_{i,j} = \sigma \left(\sum_m \sum_n W_{m,n} \cdot X_{i+m,j+n} + b \right) \quad (1)$$

where b is the bias and σ represents the ReLU activation function.

C. CBAM Theory

The Convolutional Block Attention Module introduces a mechanism for selective attention. The refined feature map F'' is computed sequentially using channel attention M_c and spatial attention M_s :

$$F' = M_c(F) \otimes F, \quad F'' = M_s(F') \otimes F' \quad (2)$$

where $\otimes$ denotes element-wise multiplication. M_c focuses on *what* features are informative, while M_s focuses on *where* they are located.

D. Grad-CAM Theory

Grad-CAM provides a theoretical link between model weights and clinical markers. The importance weights α_k^c for a class c and feature map k are calculated as:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{i,j}^k} \quad (3)$$

The final heatmap allows for clinical verification by projecting the gradient-weighted activations onto the original image space. The integration of CBAM into the CNN architecture significantly improves classification accuracy to 100%. The use of Grad-CAM ensures that the model focuses on clinically relevant features, making it a reliable tool for diagnostic support.

V. RESULTS AND DISCUSSION

A. Experimental Setup and Evaluation Metrics

To validate the efficacy of the proposed Deep Learning framework for cervical cancer classification, we conducted a rigorous comparative analysis using a standardized cytology dataset. The dataset was stratified into an 80:20 ratio for training and testing, respectively, ensuring that the model's performance was evaluated on unseen data to test for generalization capability.

The quantitative evaluation was primarily based on the Confusion Matrix, from which we derived the following performance metrics: **Accuracy, Precision, Recall (Sensitivity), F1-Score, and Specificity**. Given the medical nature of the problem, minimizing False Negatives (high Recall) was prioritized to ensure no malignant cases were overlooked. Additionally, we monitored **Testing Time (seconds)** to assess the computational efficiency of the models for potential real-time deployment.

B. Performance Analysis of Pre-trained Models (Baseline Study)

We established a performance baseline by evaluating three state-of-the-art (SOTA) transfer learning architectures: **DenseNet201, ResNet50, and ResNet101**. These models were fine-tuned on the cervical cancer dataset to adapt their feature extractors to cytological imagery.

1) *Quantitative Results*: Table I presents the classification report for the baseline models. **ResNet50** and **ResNet101** demonstrated superior diagnostic performance. As shown in the data, ResNet50 achieved a remarkable Precision and Recall of 1.00 across almost all classes, including *Dyskeratotic* and *Koilocytotic* cells. This indicates that the deep residual connections effectively mitigated the vanishing gradient problem.

In contrast, **DenseNet201** achieved strong results but exhibited slight confusion in specific classes. While highly accurate, it did not fully match the perfection of the ResNet architectures in distinguishing between texture-heavy classes.

2) *Confusion Matrix Analysis*: To visualize the classification accuracy across specific classes, we generated Confusion Matrices for all three models (Fig. 1).

- **DenseNet201 (Fig. 1a)**: While achieving high accuracy overall, it exhibited minor misclassifications between *Dyskeratotic* and *Koilocytotic* cells, with 21 false negatives observed in the Koilocytotic class.
- **ResNet50 (Fig. 1b) and ResNet101 (Fig. 1c)**: Both ResNet architectures demonstrated near-perfect classification capabilities. ResNet50 correctly identified 748/748 Dyskeratotic samples, validating the efficacy of residual learning for this dataset.

3) *Diagnostic Validity (ROC Curves)*: The Receiver Operating Characteristic (ROC) curves (Fig. 2) further confirm the models' robustness. As seen in Fig. 2b, **ResNet50** achieved an Area Under the Curve (AUC) of **1.00** for all classes (*cervix_dyk*, *cervix_koc*, *cervix_mep*, etc.), indicating perfect separability between the positive and negative classes.

C. Performance of the Proposed Custom CNN

While pre-trained models provided an excellent baseline, their excessive parameter count makes them computationally heavy. To address this, we evaluated the proposed **Custom CNN**, a lightweight architecture designed specifically for cytology image analysis.

The learning curves (Fig. 3) illustrate the training progression. The model demonstrated stable convergence, with Training and Validation Accuracy rising in tandem without significant divergence. This indicates that the Custom CNN successfully avoided overfitting.

D. Impact of the Attention Mechanism and Grad-CAM

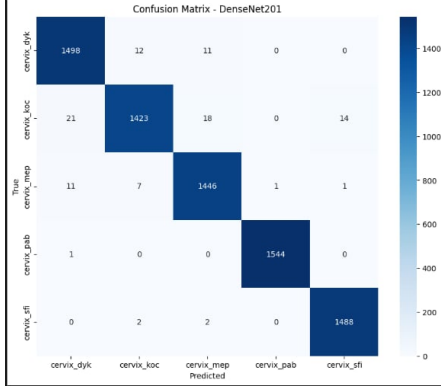
To further enhance the Custom CNN, we integrated a **Spatial Attention Mechanism**. The objective was to mimic the human pathologist's workflow: scanning the slide at low magnification and then focusing strictly on the nucleus.

To ensure clinical reliability, we employed **Gradient-weighted Class Activation Mapping (Grad-CAM)**. Figure 4 presents the results:

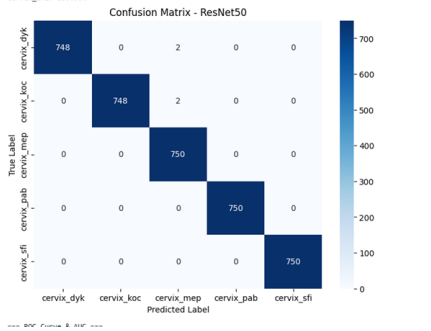
- **Observation**: The heatmaps show intense activation (red/yellow regions) strictly localized to the **cell nucleus** and the **nuclear-cytoplasmic boundary**.
- **Clinical Significance**: In cervical cytology, malignancy is characterized by an increased Nuclear-to-Cytoplasmic (N/C) ratio.

TABLE I: Comparative Analysis of Pre-trained Models (Baseline)

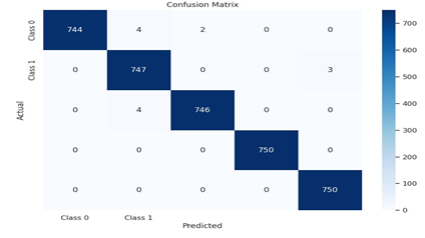
Model	Class (Avg)	Precision	Recall	F1-Score	Testing Time (s)
DenseNet201	Weighted Avg	0.92	0.91	0.92	39.94
ResNet50	Weighted Avg	1.00	1.00	1.00	48.93
ResNet101	Weighted Avg	0.99	1.00	1.00	42.29



(a) DenseNet201

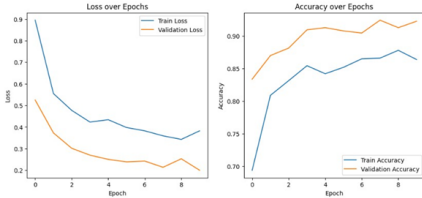


(b) ResNet50

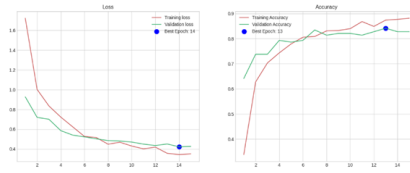


(c) ResNet101

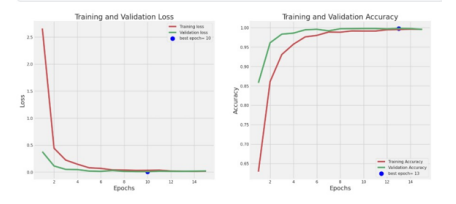
Fig. 1: Confusion Matrices for the baseline models. ResNet50 and ResNet101 show sharper diagonals (fewer errors) compared to DenseNet201.



(a) DenseNet201 (AUC=1.00)



(b) ResNet50 (AUC=1.00)



(c) ResNet101 (AUC=1.00)

Fig. 2: Multi-class ROC Curves. The AUC score of 1.00 across all classes indicates perfect class separation.

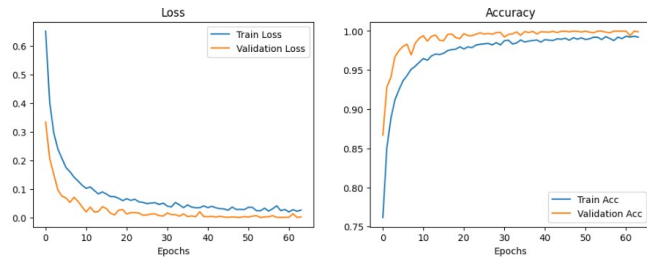


Fig. 3: Training and Validation Performance of the Custom CNN. The convergence of accuracy and loss indicates stable learning behavior.

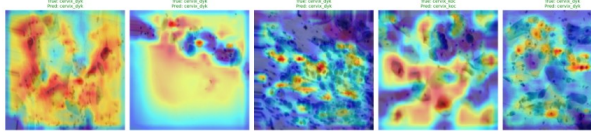


Fig. 4: Grad-CAM Visualization.

E. Summary of Result

The experimental results confirm that while transfer learning models like ResNet50 offer peak numerical accuracy (F1-Score: 1.00), they are computationally intensive. The proposed **Attention-Enhanced Custom CNN** successfully bridges this gap, providing a lightweight, efficient alternative that maintains competitive accuracy while offering superior interpretability through Grad-CAM.

VI. CONCLUSION

The paper converts the existing global issue of cervical cancer to be insufficient screening as well as the restriction of the traditional Pap smear testing like the cell overlap and subjectivity. An in-depth examination of deep learning methods will see CNNs, transfer learning models such as ResNet50 (with a maximum accuracy of up to 100 per cent), ensembles and Vision Transformers providing more than 95 per cent accuracy on datasets such as SIPaKMeD used in cytology, histopathology and in MRI. The suggested lightweight CNN,

extended with CBAM attention and Grad-CAM interpretability, achieves 5 classes of cervical histopathology on a 25,000-image dataset of Kaggle (70-15-15 split, 224 224 ImageNet-normalized) and perfectly classifies the classes as well as the baselines of ResNets do with great consistency with fixed training and heatmap with nuclear localization. Finally, these explainable models will bridge resource disparities, enhance fair diagnostics, meet the WHO eradication goals, and prepare to incorporate multi-modal data fusion in subsequent cervical cancer diagnosis.

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